

Vanadium Pentoxide (V_2O_5) Gel Battery

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Abstract

A hydrated vanadium pentoxide gel ($V_2O_5 \cdot n H_2O$) was synthesized via slow acidification of sodium metavanadate. XRD and DSC confirmed a layered structure with intercalated water. The intrinsically acidic gel enables cation exchange and was deposited on conductive substrates (SnO_2 or ITO) to create electrochromic electrodes showing pronounced voltage-induced color changes (electrochromism).

Keywords: $V_2O_5 \cdot n H_2O$ Xerogel, Sol–Gel Polymerization, Electrochromism, Ion Intercalation, SnO_2 /ITO Conductive Substrates

Introduction

The aim of this lab work is to assemble an electrochemical cell in which the active material is a vanadium pentoxide (V_2O_5) gel deposited on semiconductor substrates of SnO_2 . The first part of the lab focuses on the actual synthesis of the gel via inorganic polymerization of neutral monomers $VO(OH)_3$ through oligation and oxolation mechanisms. The next part involves the fabrication of semiconducting glass slides. We will then study the colorimetry of this gel via redox reactions, so as to visualize the real-time operation of the electrochemical cell.

1. Experimental Part

1.1 Gel synthesis - Precursor preparation

A 50 mL solution of solid $NaVO_3(s)$ at 0.25 M is prepared by heating at 60–65°C for one hour to promote dissolution. The suspension, initially a milky white at $pH \approx 8-9$, becomes a clear solution at $pH \approx 9$.

The goal is to generate the neutral precursor $VO(OH)_3$ so as to produce a vanadium gel via inorganic polymerization. To do that, the pH is lowered by acidifying the solution. We perform a controlled acidification using an ion-exchange column. We pack a glass burette with a glass wool plug at the bottom, then add approximately 10 cm of Na^+/H^+ cation-exchange resin (DOWEX 50WX-100 mesh) in the H^+ form. We rinse the column with distilled water until the effluent reaches a near-neutral pH. The solubilized solution is then added dropwise to avoid disturbing the packed resin bed. The Na^+ ions in the eluent exchange with the H^+ ions on the resin, acidifying the solution. By rinsing with distilled water, four eluate fractions of approximately 25 mL each are collected.

Initially, the solution was clear but turned colored upon acidification. This color change arises from the complex's geometry shifting from tetrahedral to octahedral due to the increased presence of water in its coordination sphere. The precursor solution is orange and it reverts to yellow in the fourth eluate. The resin also changes color during elution. Originally orange, the binding of Na^+ converts it to dark brown. Once the solution is fully acidified, the column is regenerated by passing 50 mL of 6 M HCl, then rinsed with distilled water to restore neutral pH and the resin's initial orange color. Care is taken to keep the resin moist with some distilled water.

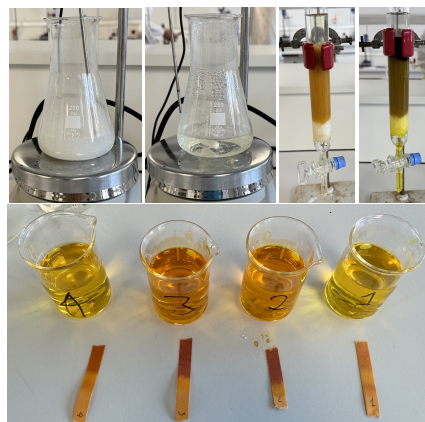


Figure 1. First, we solubilize $NaVO_3$ in distilled water (milky white solution before, clear solution after) and then we acidify it in an ion-exchange column (Na^+/H^+) (orange resin before, dark brown resin after). We extract four 25 mL eluates with gradual acidification (1 : $pH \approx 6$, 2 : $pH \approx 2-3$, 3 : $pH \approx 2-3$, 4 : $pH \approx 1-2$).

1.2 Gel synthesis of V_2O_5

Two 10 mL portions of the orange eluate are taken, and a spatula tip of KCl is added to one of them. After drying in an oven at 68–70°C, a reddish-brown gel forms via condensation mechanisms. We notice that in the fraction containing KCl, the gel is less firm, which is entirely expected. Temperature also affects the rate of gel formation and its final texture.

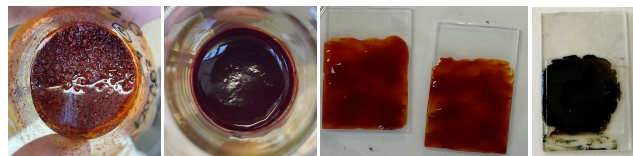


Figure 2. Synthesized gels with KCl and without KCl, that are coated on a glass slide (before and after oven drying).

1.3 Gel structure of V_2O_5

To characterize the crystalline structure of the gel, we spread a layer of wet gel approximately 2 mm thick onto three clean glass slides, then dried them in an oven at 68–70°C for 30 minutes. This process yields a fibrous xerogel of $V_2O_5 \cdot n H_2O$.

From the X-ray diffractogram, we can determine with Bragg's law the interlayer distance for dehydrated V_2O_5 and the effect of various metal cations into the gel, replacing the existing H^+ , and influencing on the gel's crystalline structure.

The DSC thermogram shows two phase transitions, one within a broad temperature range around 160.13°C , and the other sharper, occurring near 265.87°C .

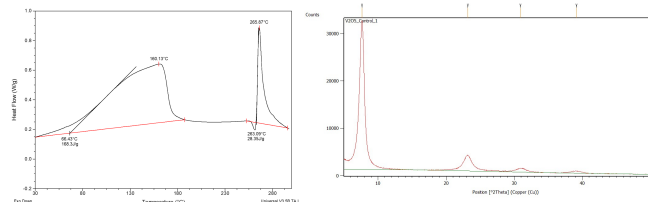


Figure 3. Left figure : DSC thermogram of the gel (visualization of two exothermic peaks at 160.13°C and 265.87°C). Right figure : X-ray diffractogram of the synthesized gel.

1.4 Preparation of the Semiconductor slides

Several glass slides are placed in an oven at 600°C for 2 min. Then a solution of $\text{SnCl}_4 \cdot 5\text{H}_2\text{O}$ in methanol is sprayed on the slides, that are then placed in the oven for 2 min. This process is repeated until a thin layer of SnO_2 is obtained. Using a multimeter, we verify that the surface is conductive. We created a *n*-doped semiconductor ! We also use commercial slides coated with ITO (Indium Tin Oxide) in the lab.

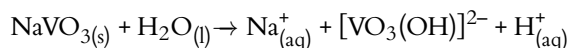
1.5 Electrochemical properties of the Gel

One of the electrodes consists of a small amount of gel on an ITO-coated slide covered by a cover slip. The other electrode is a tin oxide semiconductor slide. The electrolyte solution is 50 mL of a 1 M aqueous solution of the metal salt NaCl. The electrodes are connected using crocodile clips, which must not come into contact with the liquid. The gel electrode is connected to the negative terminal. A potential difference of 5 V is applied. An electrochromic effect is observed : the gel changes from dark red to green after about one minute of reduction. If the polarity is reversed, the gel partially regenerates as the color changes back from green to dark red in part.

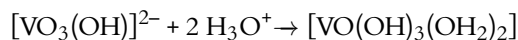
2. Results & Discussion

2.1 Gel synthesis - Precursor preparation

In water, the V^{5+} ions become hydrated, forming tetrahedral $\text{VO}_3(\text{OH})^{2-}$ complexes at $\text{pH} \approx 9$ according to the reaction :



The goal is to generate the neutral precursor $\text{VO}(\text{OH})_3$ so as to produce a vanadium gel via inorganic polymerization. To achieve this, the solution must be acidified to $\text{pH} \approx 2$:



The pH is lowered by acidifying the solution. At first glance, one might simply add a strong acid such as aqueous $\text{HCl}_{(\text{aq})}$ directly to the solution. However, this approach is too harsh and

does not yield a gel with the desired texture, as the introduced Cl^- ions can act as potential ligands and insert themselves into vanadium's coordination sphere. After adding just a few drops of acid, the solution turns deep red, then shifts to orange over a characteristic transition time on the order of one minute, corresponding to the statistical timescale for substitution of HO^- or H_2O by Cl^- . Presumably, the Cl^- ions substitute for H_2O , which are the better leaving groups. This yields a gel composed of spherical polyoxovanadate clusters [2-3]. Therefore, we instead perform a controlled acidification using an ion-exchange column. Initially, the solution was clear but turned colored upon acidification. This color change arises from the complex's geometry shifting from tetrahedral to octahedral due to the increased presence of water in its coordination sphere. Indeed, the coordination number increases from 4 to 6 when $\text{pH} < 6$, because the ligand-to-metal charge transfer (LMCT) from water molecules to vanadium decreases as pH decreases (vanadium being in the +V oxidation state). Consequently, the V - O bonds become progressively less covalent and the metal center can accommodate more ligands as the bond lengths increase, i.e. the coordination sphere expands [4]. Charge-transfer transitions are therefore higher in energy in the tetrahedral geometry than in the octahedral geometry. We therefore have :

$$\Delta E_t > \Delta E_o \Rightarrow \lambda_t < \lambda_o \Rightarrow \begin{cases} \lambda_t < 400 \text{ nm} \Rightarrow \text{transparent} \\ \lambda_o \approx 490 \text{ nm} \Rightarrow \text{dark red} \end{cases}$$

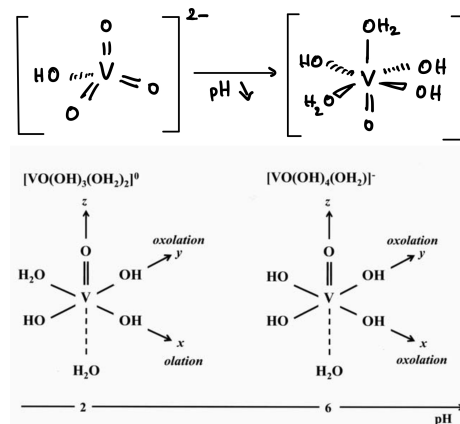


Figure 4. Tetrahedral (T_d) and octahedral (O_h) geometries of the complexes. The precursor's geometry dictates the gel's crystalline structure after condensation.

Since the absorbance of the solutions follows a cutoff-type spectrum, the neutral monomer absorbs the visible range up to $\lambda_o \approx 490 \text{ nm}$ (green), which explains why the solution appears dark red rather than pure red — because it absorbs a mixture of wavelengths. Moreover, the fact that $\lambda_t < 400 \text{ nm}$ indicates that it only absorbs in the UV, so the solution is indeed transparent. One can also note that, during passage through the column, the complex concentration decreases. That is why the precursor solution is orange instead of the expected dark red. Furthermore, after the solution has turned red, it reverts to yellow in the fourth eluate.

2.2 Gel synthesis of V_2O_5

A reddish-brown gel forms via condensation mechanisms : rapid S_N1 -type ololation and slower S_N2 -type oxolation. Since the gel network arises from V – O – V linkages, we propose that the ololation products subsequently undergo oxolation to yield the major polymer. This process builds vanadium oxide ribbons that are cross-linked by oxolation.

In the fraction containing KCl, the gel is less firm, which is entirely expected because the Cl^- ions can substitute for the H_2O ligands in vanadium's coordination sphere. This disrupts the ololation pathways (which rely on H_2O in the oxolation mechanism, along the Ox axis) and thus prevents formation of stacked-sheet gels. Spherical clusters would form if H_2O were the leaving group in the oxolation step (along the Oz axis).

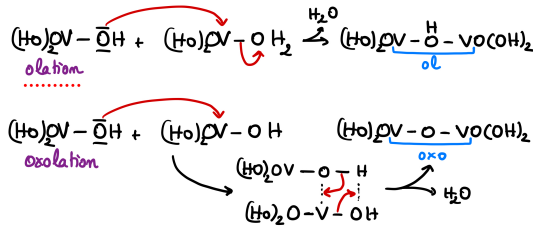


Figure 5. Inorganic condensation polymerization mechanisms.

Temperature also affects the rate of gel formation and its final texture. If it is too high, thermal agitation may break bonds within the coordination sphere or between the oxide layers, causing the gel to “melt”. If it is too low, gelation can become excessively slow. Therefore, we selected a compromise temperature of 68°C – 70°C for the drying oven.

2.3 Gel structure of V_2O_5

Upon examining the precursor's geometry, one finds that condensation reactions can occur only within a specific xy plane. In particular, ololation and oxolation occur along different axes (x or y) because one coordination site remains occupied by an H_2O ligand. Ololation links V_2O_5 square pyramids corner-to-corner, forming chains of pyramidal units. Oxolation links these pyramids edge-to-edge, building stacked sheets of square pyramids. As a result, V_2O_5 complexes crystallize into straight, layered sheets of square pyramids, separated by intercalated water layers spaced about 2.8 Å apart (the Van der Waals diameter of H_2O). The trapped water confers the gel its unique texture : mechanically rigid thanks to hydrogen bonds, yet fragile because those same H-bonds can be easily broken.

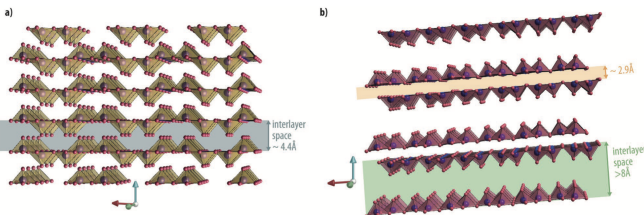


Figure 6. Structural models of V_2O_5 bilayers with and without intercalated water. Introduction of water expands the spacing to over 8 Å.

Because liquid water remains in the layers, this xerogel shows minimal diffraction peaks by X-ray diffraction (XRD), appearing XRD-amorphous in its wet state [5–8]. Oven drying yields a crystalline material whose XRD analysis allows us to determine the lattice parameter c since $d_{100} = c$. Bragg's law is given by $n\lambda = 2d_{hkl} \sin \theta$. The observed peaks in a X-ray diffractogram of a $V_2O_5 \cdot 0.5 H_2O$ gel layer correspond successively to the Bragg planes (100), (200), (300), etc. Due to the low crystallinity of $V_2O_5 \cdot n H_2O$, only a few Bragg reflections are detected. Knowing that $d_{100} = c$, and using $\lambda = 1.54 \text{ Å}$, we determine :

$$d_{hkl} = \frac{\lambda}{2 \sin \theta_{hkl}} \Rightarrow \begin{cases} d_{100} = 8.83 \text{ Å} \\ d_{200} = 4.33 \text{ Å} \\ d_{300} = 2.88 \text{ Å} \end{cases} \Rightarrow c_{V_2O_5} = 7.4 \text{ Å}$$

Experimentally, our diffractogram yields an average interlayer spacing of $\bar{c} = 11.46 \text{ Å}$, based on three control samples. The measurement uncertainty is calculated as :

$$u(c_{exp}) = \frac{1}{\sqrt{3}} \sqrt{\frac{1}{2} \sum_{i=1}^3 (c_i - \bar{c})^2} = 0.09 \text{ Å} \Rightarrow \bar{c} = 11.46 \pm 0.09 \text{ Å}$$

We deduce that the synthesized gel contains approximately $n = \frac{11.6-7.4}{2.8} \pm 0.03 = 1.50 \pm 0.03$ layers of intercalated water.

We then introduce various metal cations into the gel, replacing the existing H^+ ions to study their influence on the gel's crystalline structure. To do this, we prepare 50 mL of a 0.1 M aqueous solution of a given metal salt M^{n+} , X^{n-} . The V_2O_5 -coated glass slide is immersed in the metal cation solution for at least 10 minutes. After rinsing with distilled water, the film is dried in the oven for 15 minutes, and its XRD spectrum is recorded to obtain the following results :

Metal salt	Hydrated Ionic Radius (Å)	c_{exp} (Å)
LiCl	3.40	11.36
NaCl	2.76	11.05
RbCl	2.28	10.95
CsCl	2.28	10.90
KCl	2.32	10.99
KBr	2.32	10.93
KI	2.32	10.33
BaBr ₂		11.27
MgCl ₂	4.28	13.32
MnCl ₂		12.66
CrCl ₃	4.08	12.23
FeCl ₃		12.27

It is observed that the lattice parameter value strongly depends on the charge of the cation for a given ligand. Cations with a higher charge density tend to be surrounded by more layers of water molecules (greater solvation) to neutralize their charge, which results in a larger lattice parameter. Calculating the average lattice spacing by cationic charge for the same

ligand Cl^- gives :

Charge	+1	+2	+3
c_{avg}	11.05	12.99	12.25

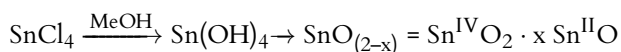
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indicates that solvation is maximal for a cation with a +II oxidation state. Next, for a fixed charge, the greater the hydrated ionic radius of the metal cation, the larger the lattice parameter. The hydrated radius increases as the bare ionic radius decreases, because a smaller ion has a more localized charge and thus attracts a larger hydration shell. Finally, for a given metal cation, the larger the counter-ion, the fewer water layers are needed for solvation due to the more diffuse ionic charge. As a result, the lattice parameter decreases.

The DSC thermogram confirms the proposed crystalline structure of the vanadium gel. The first transition corresponds to the evaporation of interlayer water molecules, leading to the formation of a more compact crystalline structure. This transition occurs above the standard boiling point of water because the water molecules are not freely mobile between the layers, making their evaporation more difficult. This process appears to be reversible : upon cooling, hydrogen bonds can readily reform between the water and the vanadium oxide sheets. The transition is characterized by a broad peak, as the water molecules are H-bonded. The second transition corresponds to the removal of water molecules from the coordination sphere of the vanadium complex. As this water is more tightly bound, the associated transition occurs at a significantly higher temperature, since the bonding energies involved in coordination complexes are much greater than those of H bonds between layers. This vaporization appears to be irreversible, as it implies the covalent bonds to break.

2.4 Preparation of the Semiconductor slides

An n -doped semiconductor is formed because the tin oxide obtained has the empirical formula $\text{SnO}_{(2-x)}$, which indicates a deficiency of oxygen in the crystal lattice. Thus, there are $2e^-$ free in the lattice because the nucleus of an oxygen atom is missing, but the crystal must remain electrically neutral.



Therefore, we have not only Sn(IV) , but also Sn(II) . The slide is transparent. Indeed, it has a band gap between the valence band and the conduction band of $E = 3.7 \text{ eV}$, so $\lambda = hc/E = 335 \text{ nm}$. Absorption is in the UV so the slide is transparent.

We also use commercial slides coated with ITO (Indium Tin Oxide) in the lab. This is an indium oxide doped with tin (In_2O_3 at 90%, SnO_2 at 10%). In practical terms, In^{3+} is replaced by Sn^{4+} , which implies having a free electron in the lattice to compensate for the extra oxidation state of the metal cation. Therefore, it is an n -doped semiconductor. The slide is also transparent because the band gap energy is close to $E = 4 \text{ eV}$, hence $\lambda = 310 \text{ nm}$. Thus, these slides are often used for their optical transparency and electrical conductivity.

Moreover, the resistances of commercial slides are almost all identical, unlike the experimental slides. Indeed, commercial slides are industrially produced under very controlled conditions (temperature, atmosphere, deposition thickness, doping rate, and material purity). This ensures a homogeneous thickness, stable composition, and uniform dopant distribution, which gives a very consistent electrical resistance from

one sample to another. Experimental slides, made manually, can show variations in deposition thickness, poor doping or structural homogeneity, contamination or partial oxidation, or deviations in annealing temperature or time. These factors strongly affect charge carrier mobility, free electron density, and thus the final resistance of the slide.

2.5 Electrochemical properties of the Gel

The gel electrode is connected to the negative terminal because we want to reduce vanadium ($\text{V}^{5+} \rightarrow \text{V}^{4+}$) :



with $x = 2$ and $n = 1.5$. A potential difference of 15 V is applied, which seems high (only about 4 V is needed for a LiCl aqueous solution), highlighting the influence of the metal cation charge and the advantage of using lithium-based electrolytes. However, solutions such as Na_2CO_3 are cheaper than Li_2CO_3 , so a compromise is needed [10]. A current of 11 mA is measured. It is observed that electrical conductivity increases as the proportion of reduced vanadium increases.

Theoretically, the amount of charge passed is : $Q = n_e \cdot \mathcal{F} = x m \mathcal{F} / M = 455x \text{ A.s.g}^{-1}$. The deposited gel mass is approximately $m = 1 \text{ g}$ and the experiment runs for 1 minute, so that $Q_{\text{exp}} = 660 \text{ A.s.g}^{-1}$ and $x = 660/455 = 1.45$. This shows that experimentally, not all the gel was reduced. The theoretical maximum is $x_{\text{max}} = 2$, so the result is consistent with the fraction of gel actually reduced.

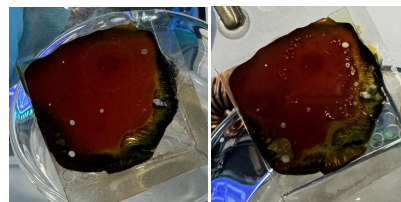


Figure 7. Gel coloration changing to green and then back to orange when reversing the connections

An electrochromic effect is observed : the gel changes from dark red to green after about one minute of reduction. Indeed, electrons flow to the negative pole and reduce vanadium within the gel. The gel remains electrically neutral because metal cations Na^+ insert into the gel structure. The gel's structure is thus composed of a mixture of vanadium ions in different oxidation states, enabling intervalence charge transfer (which occurs between metals of different oxidation states). Combined with ligand-to-metal charge transfer (LMCT), which gives the gel its dark red color, this results in the observed green coloration. If the polarity is reversed (the gel is connected to the positive terminal), the gel partially regenerates as the color changes back from green to dark red. This reversibility is not complete, and the gel is not fully restored. Additional experiments demonstrate good reversibility, around 70% after 30 charge cycles at a current density of $j = 0.05 \text{ mA.cm}^{-2}$ [10]. This result applies specifically to $\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$ bilayers, as they have a larger contact surface than simple V_2O_5 layers.

3. Environmental considerations

Due to their electrochemical properties, electrochromism, and tunable structures, vanadium pentoxide nanomaterials are widely studied for applications in energy, catalysis, sensors, and optics [11]. But what are the environmental challenges associated with this material?

1. **Abundance and production:** Vanadium is a relatively abundant metal (around 63 million tonnes worldwide). Global vanadium pentoxide production is approximately 100,000 tonnes per year. However, production is concentrated in a few countries, posing geopolitical risks (China produces two-thirds, followed by Russia and South Africa). Moreover, the potential growth of vanadium redox batteries is expected to double demand by 2030. In practice, vanadium pentoxide is obtained by acid leaching of vanadium ores or spent catalysts, reducing the need for primary extraction.
2. **Social issues:** Unlike cobalt, vanadium extraction is not associated with major human rights concerns. However, as with any metal, local environmental impacts (mine waste, water pollution) must be managed. Transitioning to aqueous vanadium batteries could reduce demand for cobalt and other critical metals, which is socially beneficial by preventing wealth concentration in a few cobalt-producing countries.
3. **Energy cost:** Aqueous solution reactions require heating to 100–200°C and long reaction times, consuming significant thermal energy. Techniques like spraying or deposition produce high-quality films but require high vacuum and temperature, resulting in high energy consumption. Some "green" methods are emerging, such as microwave- or ultrasound-assisted hydrothermal synthesis or use of biodegradable organic precursors. Nevertheless, gel synthesis remains very sensitive to growth and treatment conditions [12–14].
4. **Economic cost:** The price of V_2O_5 is moderate; in 2025, vanadium costs about 25 USD/kg. Hydrothermal or sol-gel syntheses require inexpensive equipment (ovens, standard autoclaves) but consume thermal energy. While conventional methods do not require exotic materials — limiting fixed costs — cost optimization is possible through energy efficiency and chemical recycling [12].
5. **Toxicity:** Although V_2O_5 itself is not highly dangerous, vanadium salts and dusts are respiratory irritants and can cause chronic bronchial inflammation. Vanadium ions (V^{4+}/V^{5+}) may disrupt cellular metabolism due to their ability to enter cells and generate free radicals [15].
6. **Recycling and waste management:** Vanadium is a valuable metal with established industrial recovery processes. Recycling pathways for lithium-ion batteries could be extended to vanadium-based batteries given the metal's economic importance. In case of accidents or end-of-life, V_2O_5 residues must be treated as heavy metal waste to prevent environmental release and promote recovery.

Conclusion

In this study, a hydrated vanadium pentoxide gel ($V_2O_5 \cdot n H_2O$) was synthesized via controlled acidification, yielding a layered structure confirmed by XRD and DSC. The gel's unique texture arises from intercalated water and hydrogen bonding, which also enables ion exchange. When deposited on conductive substrates (SnO_2 or ITO), the gel acts as an electrochromic electrode, exhibiting reversible redox-driven color changes. Structural modifications were observed upon intercalation of various metal cations, influencing the interlayer spacing. The synthesis method, based on sol-gel chemistry, offers tunability though still sensitive to synthesis conditions. Environmentally, vanadium is abundant but its processing and toxicity raise sustainability concerns. Advances in recycling and "green" synthesis could improve its industrial viability. Overall, vanadium gels have potential for applications in energy storage and smart materials.

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